

The paper proposes an object-conditioned framework for haptic teleoperation, where detected object classes are mapped to predefined strategies that jointly configure slave impedance, master haptic feedback, and gripper parameters. The system is evaluated through a five-mode human-in-the-loop experiment. This paper is well-organized and easy to follow. The experiments and data analysis are also well conducted. However, I have several concerns about whether the experiments sufficiently demonstrate the benefit of the proposed multi-channel adaptation.

- The current experiments do not clearly show that the full multi-channel adaptation is necessary, which is the most important motivation of this paper. The most important missing comparison is a gripper-only adaptive mode. In the current setup, object classification directly changes the requested grasp force F_g and the gripper speed v_g . For this task, it is possible that simply using object classification to adjust F_g and v_g , while keeping the master haptic and slave impedance parameters fixed, may already provide most of the performance improvement. I think an additional gripper-only or impedance-plus-gripper condition is needed to show that the proposed multi-channel adaptation really gives an extra benefit.

- One follow-up question is that why you need haptic feedback on the master side. As it is only a grasp task, displaying the kinesthetic (force/torque) feedback to the user seems not very relevant for improving the grasping task performance, unless the collision or loading/mass takes a roll. More specifically, the master-side force feedback is based on the Franka manipulator external-force estimate. It is therefore mainly kinesthetic feedback from arm/environment contact, not grasp-force feedback from the gripper fingers. The operator does not directly feel the actual grasp force. Because the main motivation of the paper is handling fragile or deformable objects, it is not fully clear how changing K_f helps with grasping these objects. The paper should explain more clearly what task information is provided by the master haptic feedback and why the selected values of K_f are useful.

- Again, the individual effects of K_t , K_r , and K_f are still unclear. The paper assumes that object-dependent impedance and haptic gains are useful, but the experiments do not really isolate their contribution. In particular, the paper should provide stronger evidence that changing K_t , K_r , and K_f actually improves task performance, rather than only changing the interaction feeling. A more direct ablation would help to show whether these teleoperation parameters are really important for the grasp-and-transfer task. It is suggested that the authors show/compare the slave control force w_c (eq. 3) and master force u_h^{base} (eq. (6)) for free-space movement, object contacting/grasping, and translation phases for difference experimental modes A to E. This will help to understand how the K_t , K_r , and K_f affect the task performance as well as the user experience.

- In addition, task completion time may be directly affected by the gripper speed v_g . The main performance metric is total task completion time, but v_g is one of the parameters that changes between strategies. Since the task includes grasping and releasing, the gripper execution speed itself can directly change the measured task time. The authors should separate or compensate for this effect, for example by reporting phase-wise timing for approach, grasp, transfer, and release. Otherwise, it is difficult to know how much of the reported time improvement comes from better teleoperation and how much comes from the gripper execution settings.

- The C-E comparison does not represent exactly the same intervention for all strategies. Some strategy parameters are identical to the baseline values. For the balanced strategy, the impedance parameters are already equal to the baseline, while for the stability-priority strategy the gripper parameters are equal to the baseline. Therefore, the actual difference between modes C and E changes

depending on the object category. For some objects, C–E mainly changes haptic and gripper parameters, while for others it mainly changes the haptic parameters only. This makes the pooled C–E result difficult to interpret as one common “multi-channel effect.” I think this point should be discussed more carefully, or additional strategy-wise analysis should be provided.

-The supervisory teleoperation and haptic loop runs at 200 Hz, but the paper does not explain why this rate was selected. The Franka control interface and Omega.7 haptic device can support much higher update rates, e.g. Franka arm for 1khz and the Omega.7 for even upto 4khz. A higher control rate can generally provide better haptic rendering and may also improve stability margins, especially when using larger stiffness or force-feedback gains such as K_t or K_f . The authors should clarify whether 200 Hz is only the application-level supervisory rate or the actual bilateral haptic update rate, and discuss how the selected frequency affects haptic quality and stability.

-The paper changes slave impedance and master force-feedback gains in a bilateral teleoperation loop, but no formal stability or passivity analysis is provided. Since the parameter sets are switched according to the object strategy, the authors should at least discuss how closed-loop stability is ensured under different interaction conditions.